Md Rayhan Ali

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Portfolio LinkedIn GitHub

Introduction

Computer Science graduate focused on full-stack development and AI/ML engineering, with hands-on experience building web applications and AI-powered software using modern Python and JavaScript frameworks. Experienced in developing RESTful backends, responsive React/Next.js applications, authentication systems, database-driven applications, and integrating AI capabilities into production-oriented software.

Education

Rajshahi University of Engineering and Technology May 2024
Bachelor of Science in Computer Science and Engineering *Rajshahi, Bangladesh*

- **CGPA:** 3.11 / 4.00
- **Relevant Coursework:** Data Structures and Algorithms (C++), OOP (C++, Java), Artificial Intelligence (Python), Neural Networks and Fuzzy Systems (Python), Digital Image Processing (Python), Linear Algebra with Computational Applications.

Work Experience

Outlier AI November 2024 – Present
Independent Contractor (Contributor → Auditor) *Remote*

- Promoted from Contributor to Auditor for consistent task accuracy and performance.
- Audited and refined AI-generated responses to ensure factual correctness, clarity, and linguistic precision.
- Provided quality feedback to improve data annotation standards and model reliability.

Young Learners' Research Lab May 2024 – December 2024
Research Assistant *Rajshahi University of Engineering & Technology (RUET), Bangladesh*

- Contributed to research activities involving data analysis, experiment setup, and literature review.
- Collaborated with faculty and student researchers to ensure methodological consistency and data integrity.

Technical Skills

Programming	Python, C, C++, Java, JavaScript, TypeScript
AI/ML	PyTorch, TensorFlow, Keras, Scikit-learn, Machine Learning, Deep Learning, NLP, Computer Vision, RAG, FAISS, BM25, LangChain, LangGraph
Data Science	Pandas, NumPy, Matplotlib, Seaborn
Web/Backend	FastAPI, Django, Django REST Framework, React, Next.js, Tailwind CSS, Zustand, REST APIs, JWT Authentication
Databases	SQL, MySQL, PostgreSQL
Tools	Git, GitHub

Projects

Company Policy Assistant – Evaluation-Driven RAG Knowledge Assistant (**GitHub**)
Python, FastAPI, Next.js, FAISS, BM25, Reranking

- Built a citation-backed RAG assistant using hybrid FAISS + BM25 retrieval, Reciprocal Rank Fusion (RRF), and cross-encoder reranking, improving Recall@5 from 55.1% to 81.0% and MRR from 48.5% to 71.7%.
- Designed a custom 39-question evaluation benchmark and LLM judge, achieving 92.3% faithfulness and 82.1% correctness.

TaskLens – Full-Stack Task Management & Image Annotation Platform (**Frontend** | **Backend** | **Live Demo**)
Next.js, TypeScript, Django, PostgreSQL, Tailwind CSS

- Built a full-stack task platform with JWT authentication, RESTful APIs, and Kanban-style drag-and-drop using Next.js and Django REST Framework.

- Implemented polygon image annotation with reusable React components and Zustand for state management.

Snipz – Chrome Screenshot & Annotation Extension (**GitHub**)
JavaScript, Chrome Extension MV3, Canvas API

- Built a Manifest V3 Chrome extension for capturing, editing, and annotating screenshots directly in the browser.
- Implemented a Canvas-based editor with freehand drawing, shape tools, undo/redo, and Chrome API integration for clipboard and downloads.

Foundations of Neural Network Algorithms
Python, LaTeX

- Implemented KNN, Perceptrons, Kohonen SOMs, and Hopfield Networks from scratch; applied them to the Breast Cancer dataset for classification.

Meal Management System (**GitHub**)
Java, Java Swing, MySQL

- Built a Java Swing app for hostel meal tracking, billing, and MySQL-backed data management, with admin authentication and automated monthly closing to reduce manual calculation errors.

Publications

A Hybrid 3D-2D Convolutional Neural Network with Spatial Attention for Enhanced Spectral-Spatial Classification of Hyperspectral Images (*Published at ICCIT 2024*)

DOI: 10.1109/ICCIT64611.2024.11022565 (*Undergraduate Thesis*)

- Designed and implemented a hybrid 3D-2D Convolutional Neural Network (CNN) with a spatial attention mechanism for hyperspectral image (HSI) classification.
- Utilized Z-score normalization and Incremental Principal Component Analysis (IPCA) for efficient dimensionality reduction, optimizing computational resources and enhancing spectral-spatial feature extraction.

Achievements

- Participated in RUET GanJam Intra RUET programming contest — 2020.
- Received Dutch Bangla Bank Scholarship for Undergraduates — 2019–2023.
- Received Technical Scholarship from RUET based on Admission Merit — 2019, 2020, 2022.

References

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